

Varun Mahankali

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Professional Summary

Software Engineer with 3+ years of experience designing, developing, and deploying API-driven applications using Angular, React, Next.js, and PostgreSQL. Recently completed hands-on training in Generative AI, with practical experience building Retrieval-Augmented Generation (RAG) pipelines and LLM-based inference workflows through projects and coursework. Strong background in backend system design, data processing, production deployments, and collaboration in Agile/CI/CD environments to deliver scalable, secure, enterprise-grade applications.

Technical Skills

- **Programming Languages:** C++, SQL, Python, Java, JavaScript
- **API & Backend:** RESTful API Design, JSON, Secure Services, Backend System Design, Node.js, Next.js
- **Web Technologies:** HTML, Tailwind CSS, JavaScript, Angular, React, Bootstrap, TypeScript
- **Databases:** MySQL, MongoDB (Foundational Relational & NoSQL Concepts), Vector Databases, NoSQL, PostgreSQL
- **GenAI Frameworks & Tools:** LangChain, Hugging Face Transformers, Embeddings, Vector Search, Tokenization
- **AI/ML Foundations:** NLP Fundamentals, Feature Engineering (Foundational), Model Inference, Text Embeddings, Similarity Search
- **Tools & Platforms:** Git, GitHub, Postman, Visual Studio, Gitlab, Tableau, Databricks, PowerBI, Swagger, AWS, Microsoft Office
- **SDLC & Practices:** Agile/Scrum, CI/CD, Code Reviews, Production Support, Secure Coding
- **Core Foundations:** Data Structures, Algorithms, Object-Oriented Design, System Design

Experience

Micro1 Inc.

May 2026 - Present

AI Trainer

Remote

- Train and evaluate Large Language Models (LLMs) by designing prompts, checking outputs, and improving response accuracy.
- Partnered with engineering and AI teams to refine model accuracy and context management, achieving a 30% reduction in errors and improving response reliability.
- Validate and analyze AI outputs using Python and libraries like pandas and NumPy, proactively identifying errors and implementing solutions that consistently improve model accuracy and reliability.

Kalven Technologies

Jul 2025 - Mar 2026

Software Developer

Dallas, Texas

- Designed and developed **RESTful APIs** with **NodeJs, React**, enabling enterprise data platforms to process **15%** more transactions per day
- Performed data cleaning, validation, and integrity checks using Python scripts, ensuring **99%** accuracy of incoming and outgoing datasets
- Built web-based user interfaces with JavaScript, HTML, and CSS, reducing user task completion time by **20%**
- **Debugged, reviewed, and optimized code**, improving system performance by **30%** and increasing code maintainability
- Implemented end-to-end data pipelines in Python that transformed raw data into structured JSON outputs for internal analytics, cutting report generation time in half
- Collaborated in an **Agile/Scrum team**, using **Jira** to plan sprints, attend daily **stand-ups** and retrospectives, and implement process improvements that streamlined sprint planning and enhanced team coordination
- Supported **production deployments**, monitored system health, and resolved operational issues to maintain application resiliency.
- Followed **OWASP secure coding standards** and SDLC processes, meeting internal compliance requirements and reducing security vulnerabilities in released code

eGlobal Doctors

Aug 2023 - May 2025

Software Engineer

Dallas, Texas

- Developed and maintained API-driven web applications with **JavaScript, Next.js, and Angular**, using **Git for version control** and Postman for testing, which lowered API error rates and ensured reliable service delivery
- Designed backend services for scheduling, payments, and account management with a focus on security and scalability, supporting a **30%** increase in user traffic while reducing payment errors by **20%**.
- **Lead and mentor junior developers on multilingual feature development**, conduct code reviews using GitLab pull-request tools, and improve code quality, which speeds up onboarding and reduces review cycle time

- **Leveraged AWS database services** to optimize SQL performance through indexing and query tuning, reducing report generation time and improving data accessibility for internal teams.
- Built user-facing interfaces that connected to backend services, increasing user task completion speed by **25%**.
- Participate in **code reviews** to identify defects, performance bottlenecks, and security gaps, applying **secure services** knowledge, which raises code quality and reduces production issues
- Supported **CI/CD pipelines** and deployment workflows, helping reduce deployment errors by **20%** and improve release reliability.
- Worked with **product, QA, and operations teams** to deliver stable enterprise applications, reducing issues and downtime.

eGlobalDoctors

Apr 2023 - Jul 2023

Software Engineer Intern

Hyderabad, India

- Assisted in developing and testing **API-driven web application features** using JavaScript, ensuring smooth data flow between frontend and backend components.
- Collaborated with senior engineers to **debug, optimize, and enhance existing modules**, improving application stability and performance.
- Built and maintained database and backend services with **Next.js, REST APIs, and GraphQL**, enabling reliable scheduling, user management, and data handling, which reduced data errors and improved system reliability
- Gained hands-on experience with **Git-based version control, code reviews, and Agile development practices**, contributing to sprint-based deliverables.

Projects

GenAI RAG-Based Chatbot

LangChain & Hugging Face

- Designed and implemented a Retrieval-Augmented Generation (RAG) chatbot using LangChain and Hugging Face transformer models.
- Built document ingestion and preprocessing pipelines, generating text embeddings and enabling vector-based similarity search.
- Implemented prompt chaining and context management to improve response relevance and reduce hallucinations.
- Integrated the RAG pipeline into a backend service to support scalable inference and extensible knowledge sources.
- Evaluated response quality and optimized retrieval parameters for improved accuracy and performance.

Eagle

Alumni Networking Platform

- Designed and optimized RESTful APIs supporting role-based access control and event management workflows.
- Performed data consistency checks and optimized database queries to improve response times and scalability.
- Developed web interfaces enabling seamless interaction with backend services.

Patient Obligation Perceiver Using Eye Gestures

- Contributed to application logic, data processing pipelines, and system validation.
- Integrated real-time data processing to improve system accuracy and reliability.
- Co-authored a research paper and contributed to a granted Patent.

Education

University of North Texas

Aug 2023 - May 2025

Master of Science, Computer Science

Denton, TX

- **GPA:** 3.82/4
- **Achievements:** Upskilled during Master's coursework by applying theoretical concepts to hands-on software engineering and data-driven projects, Developed system-level programs using C/C++
- **Coursework:** Completed advanced graduate coursework in Algorithms, Operating Systems, Machine Learning, and Data Science

VNR VJIE

Aug 2019 - May 2023

Bachelors

Hyderabad, India

- **GPA:** 8.7/10
- **Achievements:** Published two research papers , Mentored juniors for MERN stack course, Improved Leadership skills
- **Coursework:** Completed foundational undergraduate coursework in Data Structures, Algorithms, Operating Systems, Database Systems, and Software Engineering

Certifications & Achievements

- **AWS Certified Developer – Associate:**2025
- **Patent Holder:**“Patient Obligation Perceiver Using Eye Gestures” (IEEE, 2023)
- **Completed course:**Generative AI using LangChain and Hugging Face